

# Lectures

## Stat 151A: Linear Models

Lectures marked with a \* are still in progress.

These lecture notes are meant to supplement the reading indicated in the course schedule.

### Introduction

- [Course philosophy](#)
- [Sample means](#)
- [Linear algebra review](#)

### Unit 1: Multilinear regression

- [Simple linear regression](#)
- [Multilinear regression](#)
- [Paths to least squares](#)
- Transforming the regressors
  - [One-hot encodings](#)
  - [Interacting continuous regressors](#)
  - [Supplement: Linear transformations](#)
- [Transforming the response variable](#)

### Unit 2: Inference for linear regression

- [Confidence intervals \(review\)](#)
- [Modeling of OLS](#)
- [OLS under normality](#)

- Supplement: The multivariate normal
  - Supplement: The distribution of the residual variance estimator
  - Variable selection and F-tests
- OLS under non-normality
  - Supplement: Asymptotics under misspecification
- Statistical versus practical significance

### **Unit 3: Machine learning and regularization**

- OLS for prediction with expressive basis functions
- Bayesian methods and L2 regularization
- Sparsity and L1 regularization